# An internal orbit with deficit two

Human Compression project: L044 continuation

Let $G = \langle \rho_0, \dots, \rho_{r-1} \rangle \le \mathrm{Alt}(\Omega)$ be transitive with a connected string C-marking and all maximal omissions intransitive. Assume $n = 2r + 1$ and the actual transitive six-point head on $D = \{1, \dots, 6\}$, with the literal interface

$$\tau = \rho_3 = (6\ b)\delta, \quad R = G_{\ge 3} = \mathrm{Alt}(\Omega \setminus \{1, \dots, 5\}), \quad \sigma = \rho_4, \quad \sigma|_D = 1.$$

The involution $\delta$ fixes $D, b$. Put $H = G_{\ge 5}$ and $K = G_{>5}$, and suppose $H$ fixes exactly $D \cup \{b\}$. All eligible edges below are eligible for the full ambient omissions. Use actual normalized assignments of two such edges of each tail colour to a single $H$-orbit.

**Theorem 1.** *Suppose the two nontrivial $H$-orbits are $V$ of size $2s$ and $O$ of size $2q$, where $s \ge 2$ and $q \ge 3$ is odd. Assume $\delta$ preserves $O$ and is free there, that $K$ is intransitive on $O$, and that*

$$5 \in F_V, \qquad |F_V| = s, \qquad |F_O| = q - 2.$$

*Then this configuration is impossible.*

*Proof.* The acquired actual head-overlap argument gives

$$x = b^\sigma \ne b, \qquad \delta x = x, \qquad Kx = x. \tag{1}$$

These conclusions use natural $R$ and ambient IP, without requiring $G_{\ge 4}$ transitive. The point $x$ lies in $V$. Since $\delta$ centralizes $H$ and fixes $x$, it fixes all of $V$. The complete selected-colour equality makes $G_{F_V}$ transitive on $V$. The acquired fixed-anchor lemma then gives

$$F_V = \{5, \dots, a-1\}, \quad F_O = \{a, \dots, r-1\}, \quad a = s + 5 \ge 7, \quad r = s + q + 3. \tag{2}$$

Set $T = G_{F_O}$ and $P = G_{\le a-1}$.

We first record two literal permutation facts needed to justify projection. The prefix $P$ contains the pure group $\mathrm{Alt}(V)$. Here is the degree-independent part of the acquired prefix-kernel proof. Conjugate $\tau$ by the transitive head and multiply two conjugates; the exterior $\delta$ cancels, supplying the three-cycles of the pure $\mathrm{Alt}(D \cup \{b\})$. Conjugation by $\sigma$ merges that support with $D \cup \{x\}$ and supplies $\mathrm{Alt}(D \cup \{b, x\})$. Inductively a new mark fixes the earlier support two indices back, because it centralizes its natural alternating group. The two current supports therefore overlap in at least seven points. The complete support-merging lemma gives the pure alternating group on the enlarged actual head orbit. The head orbit of $P$ contains $x$ and all its $G_{F_V}$-translates, hence all of $V$. This proves the asserted inclusion. No even-degree budget or rank induction enters these steps.

Secondly, every mark of $T$ except possibly $\rho_a$ fixes $V$ pointwise. It fixes $x$ by (1) and commutes with the transitive group $G_{F_V}$ by (2). Thus all $T$-orbits in $V$ have size at most two. All its marks also fix $D, b$.

Project $O$ onto the $q$ pairs of $\delta$. Select one eligible edge for each colour in $F_O$. Their projections form a forest $\mathcal{F}$ with $q - 2$ edges and two components: an alternative projected path lifts using $\tau$ and the other selected colours, contradicting ambient eligibility. This is the complete pair-forest argument, before any equality conclusion is used. Each component lies inside a $T$-orbit on pairs. Consequently that action has either one or two orbits.

The pair image of $\rho_5$ is trivial. Indeed $\rho_5$ commutes with $T$. If $T$ is transitive on pairs, an involution of the odd set of pairs fixes a point, and a centralizing permutation fixing one point fixes them all. Otherwise its two orbits are the two forest components; their sizes are unequal since $q$ is odd, so $\rho_5$ preserves each. If it moved a vertex to another vertex in the same component, join them by the unique selected tree path. Any edge of that path, of colour $j \neq 5$, would acquire an alternative path using $\rho_5$, the other tree edges and $\tau$, all in $G_j$. This is again impossible.

Since $H$ is transitive on $O$, its pair action is transitive, and the same now holds for $K$. Its assumed intransitivity on points gives exactly two $K$-orbits of size $q$, each meeting each pair once. Any element of $K$ with trivial pair image fixes both rows pointwise. In particular the kernel of the pair projection on $T$ fixes $O$. It fixes $D, b$ as well, and hence is a pure even permutation of $V$. The kernel therefore lies in $\mathrm{Alt}(V) \leq P$. Actual ambient IP gives $T \cap P = 1$, so the projection on $T$ is injective. The projected consecutive marking thus retains both IP and all its actual noncommutations.

Suppose first that $T$ is transitive on pairs. Its point orbits in $O$ are then the two rows of size $q$. The mark $\sigma$ centralizes $T$ and must carry a $T$-orbit to one of the same size. Outside $O$ its orbits have size at most two, whereas $q \geq 3$. Thus $\sigma$ preserves $O$. So do $\tau$ and $H$, contrary to the natural transitivity of $R$.

Hence $T$ has exactly the two forest components as pair orbits. Every projected mark is precisely its selected transposition. To see this, an additional edge lies within a forest component. If its tree path contains an edge of a different colour, it gives the same forbidden alternative path. A path using only its own colour consists of the single selected edge. There are no other possibilities. The actual marking is connected and its projection is faithful. Thus consecutive projected marks do not commute, so all selected edges belong to one component, and the other is a singleton. The tree component is a path: three incident edges would require three distinct indices to be pairwise consecutive. Its successive indices are consecutive in order. We have proved

$$T \text{ on pairs is the natural } S_{q-1} \text{ and one fixed pair.} \tag{3}$$

On $O$, it has two orbits of size $q-1$ and two fixed points, using the faithful rows of $K$.

The permutation $\sigma$ cannot preserve $O$, since $\tau, H$ preserve it and generate $R$ together with $\sigma$. As $\sigma$ centralizes $K$, a point crossing out of $O$ brings an entire $K$-row of size $q$ to a $K$-orbit outside $O$, necessarily in $V$. On that row $T$ has an orbit of size $q-1$, by (3). All its orbits in $V$ have size at most two. It follows that $q-1 \leq 2$, and therefore $q = 3$.

Now $a = r-1$, $T = \langle t \rangle$ with $t = \rho_a$, and its pair image is one transposition of three pairs. The transitive pair image of $K$ contains that transposition, and is therefore $S_3$. Put $u = \rho_{a-1}$. For $6 \leq i \leq a-2$, commutation with $t$ shows that the pair image of $\rho_i$ is either the identity or the image of $t$. In the second case $\rho_i t$ fixes $O$, by faithfulness of the literal $K$-action on its two rows. It is then a pure even permutation of $V$, hence belongs to $\mathrm{Alt}(V) \leq P$. Since $\rho_i \in P$, this would give $t \in P$, contradicting actual independence. Consequently every such $\rho_i$ fixes $O$ pointwise. The two remaining images, those of $u, t$, generate $S_3$, so they are distinct transpositions. The group $Q = \langle u, t \rangle$ thus has exactly two natural three-point orbits on $O$. We make no assertion that its whole action outside $O$ is faithful.

For $4 \leq l \leq a-3$, put $Z_l = O^{G_{\leq l}}$, the union of the indicated translates. The prefix centralizes $Q$, so $Z_l$ is a union of actual $Q$-orbits of size three. The group $Q$ fixes $D, b, x$, eight points disjoint from every $Z_l$. Since $\sigma$ moves a $Q$-orbit out of $O$, $|Z_4| \geq 9$. For $4 \leq l \leq a-4$, every mark of index at least $l+2$, other than $u, t$, fixes $O$ pointwise and commutes with $G_{\leq l}$; it therefore fixes $Z_l$ pointwise. The group $Q$ preserves $Z_l$. If $\rho_{l+1}$ also preserved $Z_l$, then all of $G$ would preserve that proper nonempty set. Transitivity therefore forces enlargement, and commutation with $Q$ makes the increment at least one whole orbit of size three. We obtain

$$|Z_{a-3}| \geq 9 + 3(a-7) = 3a - 12, \qquad n \geq (3a-12) + 8 = 3a - 4. \tag{4}$$

On the other hand $r = a+1$ gives $n = 2a+3$. Thus $a \leq 7$, and (2) forces $a = 7$ and $s = 2$. But $F_V = \{5, 6\}$ now assigns two distinct 6-edges to the four-point set $V$. An involution with those edges fixes no point of $V$, contrary to $\rho_6 x = x$. This final contradiction proves the theorem. $\square$

**Attribution and remaining scope.** The actual head interface, selected-colour equality, fixed-anchor lemma, support-merging kernel proof and pair-forest argument are acquired inputs. The new steps are the faithful terminal projection through the actual pure kernel, the two-component analysis, and the final three-point orbit growth. Root's earlier support-kernel and orbit-matching contributions are retained in the acquired material. There is no new computation, rank induction or formal premise. This excludes the stated normalized budget only; split-orbit assignments, global attachment and the complete L044 recovery remain separate. No original, question, formal or numerical credit is claimed.